

the CPU can control the other two components, the VU1 and GS are equals, and communicate through the GIF. As mentioned above, the VU1 keeps generating the display list and the GS turns them into bitmap images. The GS has a 150 MHz core with 16 pixel pipelines. Using these pipelines, it can churn out 75 million polygons and 2.4 billion pixels per second. Compared to current GPUs, such as the GeForce FX, this may seem meagre, but considering that the PS2 was released in 2000, this is impressive.

No doubt, the PS2 is a powerful machine, but it may not be better than the latest PC, thanks to companies such as Intel and nVidia. However, it is still a good buy.

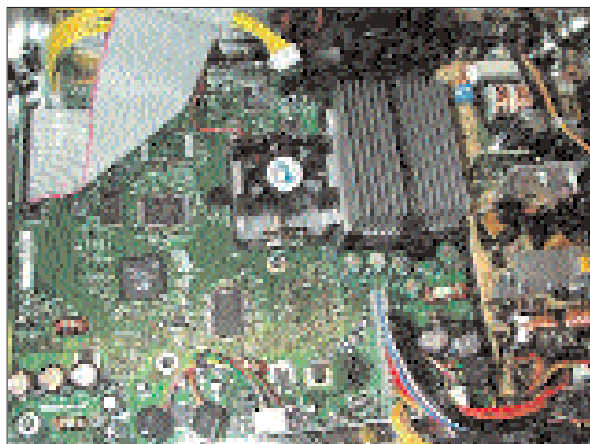
XBox: The black beauty

Keeping up with its tradition of entering every relevant market, Microsoft entered the gaming arena with its famous Xbox in 2001. The console is made of PC components such as a Pentium III 733Mhz (Coppermine), nForce chipset-based motherboard, 64 MB DDR RAM, 8 GB hard disk, a DVD drive, and runs on a stripped down version of Windows 2000. But that is where the similarity between a PC and the Xbox ends. There is a tighter integration between these components that enables it to outperform a PC. Strictly speaking, the components are not exactly the same as PC components. Let's take a look at each of the components in detail.

The Coppermine processor comes with a 128 KB L2 cache memory. Although this makes it look like a Celeron, it's definitely not one. This is because the Coppermine core here has an 8-way set associative cache rather than the 4-way cache found in the Celeron. This makes it a Pentium III with half of its cache disabled. But the lingering question is, in the days of gigahertz wars, how does a 733 MHz processor compete? The answer to that is simple—even this much processing power is not required, at least for now.

The graphics solution provided by nVidia is similar to its nForce chipset with minor variations. Like the nForce, the Xbox has the Integrated Graphics Processor (IGP) as the northbridge. The IGP acts as both the northbridge and the GPU. The difference is that the nForce

has a GeForce 2MX (NV11) class GPU, and the Xbox has an NV2A core. The NV2A is something that lies between the GeForce 3 (NV20) and the GeForce 4 (NV25). It takes care of all the graphics processing; it provides hardware transform and lighting (T&L), something which the PS2 sorely lacks. In fact, the GS of the PS2 is much inferior to the NV2A, which has two Vertex Shaders—a huge benefit because a Vertex Shader is used heavily by current generation games. The strength of the GS lies in its parallel computation power and not in the functions that it provides. Like the GS, the NV2A takes care of all the graphics processing. So the CPU is left with AI, physics and managing the connection between individual elements, such as the hard disk. For performing these functions a Coppermine is overpowered as all the



Take a peek under the hood of Xbox

games are rendered at a resolution of 640 x 480 x 32, which is sufficient for NTSC television. Xbox also supports HDTV output, for which a Conexant video encoder chip increases the basic resolution.

The southbridge is the Media & Communications Processor (MCP), which provides the audio solution. An interesting point is that a HyperTransport bus, which is used by AMD, connects the MCP and IGP. Audio is another area where the Xbox is much better than the PS2. The MCP provides a Dolby Digital 5.1 audio output, compared to the stereo output of the PS2. This is a very important feature, as many games support this surround sound format. The MCP also offers a 10/100 Ethernet controller, which allows multiple Xbox consoles to be interconnected.

Another reason the Xbox produces stunning visuals is that developers heav-

ily optimise the games for it. This is due to the fact that the system is not going to change, unlike a PC. While programming for the PC, no sane developer will develop a game that will run only on a top-of-the-line system. Any game that is released, now will be programmed to run on any card, ranging from a lower end graphics card, such as the GeForce 2 class, to the latest cards such as the Radeon 9800. This prevents the developer from optimising the game for a specific system. Also, unlike the PS2, the Xbox is closely related to the PC, which makes it a friendly turf for PC game developers.

The end and future...XBox 2, PS 3

Microsoft and Sony are busy gearing up their warriors for the future. Not much is known about the Xbox 2. After speculations that Microsoft has opted for ATI, there is a new round of rumours that Microsoft may design the processors on its own—it has apparently licensed some SGI patents. This would enable the company to strategically price the Xbox 2. Also, the CPU for the Xbox 2 is rumoured to have the ability to decode and execute instructions in Microsoft Intermediate Language (MSIL)—an integral part of .Net, while being compatible with Xbox at the same time.

Now, coming to Sony, the word that comes to everyone's mind when the subject of PS3 is brought up is the 'Cell'. Dubbed as a supercomputer on a chip, the Cell is the microprocessor that will power PS3 when it comes out, some time in 2005. Sony, Toshiba and IBM are developing it jointly. It will be fabricated using the 0.10-micron silicon-on-insulator (SOI) process technology. The PS3 will also make use of Grid computing. Grid computing is a variation of distributed computing, and presumably involves networked game machines sharing software, processing power and data.

On the whole, gaming consoles are the better option in the long run. After all, they are designed for gaming right from scratch. This has helped the industry understand the basic requirements of 3D and multimedia applications. Without a doubt, this will affect the designs of PCs in the future and make them gamer friendly. ■

MOULY ARUN PRABHU
(mouly_arunprabhu@thinkdigit.com)